

Anil Kumar

Machine Learning Engineer—Computer Vision & AI Specialist

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in anilcs13m
8+ years in CV & DL

Summary

AI Engineer with 8+ years of experience in Computer Vision & Multimodal AI, delivering production-ready solutions across image, video, and text domains, with expertise spanning 70% in large-scale data curation and 30% in model training-to-deployment, and maintenance.

Education

M.Tech in CS	Indian Institute Of Technology—Madras	2013–2015
B.Tech in CS	Gautam Buddha Technical University-Lucknow	2007–2011

Professional Experience

Jan 2020 – Present **ZEE Entertainment Enterprises – Software Engineer III**

- Led the design and deployment of large-scale AI solutions for video understanding, content moderation, video engagement, with ownership from data curation to production deployment.
- Designed and deployed a video moderation pipeline using object detection, OCR, NSFW classification (I3D + Transformer Encoder), and audio analysis — automated 70% of unsafe video detection, reducing manual review workload by 60%.
 - Built an AI-powered thumbnail generation system leveraging face detection, NIMA scoring, emotion recognition - boosted click-through rates by 15%.
 - Developed a video engagement prediction model leveraging I3D and VGGish features -improving cold-start recommendation accuracy for new videos by 20% .
 - Built contextual ad injection system by video understanding, face recognition, visual similarity (FAISS), knowledge graphs, and LLM reasoning — increased ad relevance and monetization ROI.
 - Developed "Movie X-Ray" features and scene-level search using face recognition, person tracking, and LLM-based scene-to-text conversion — enhancing movie understanding and improving OTT user experience.

Feb 2018 – Dec 2019 **Charmboard – Software Engineer**

- Contributed to building AI-powered fashion discovery and shoppable video solutions, focusing on visual similarity, video processing, and product search at scale.
- Developed a visual similarity search system for clothing and accessories using CNNs, triplet loss, FAISS, and object detection — improved image-to-image and text-to-image search relevance 70%.
 - Built shoppable video features enabling "shop the look" through video injection powered by object detection, person tracking, face detection/recognition, and visual similarity — increased user engagement, shoppable purchases and boosted product click-through rate.
 - Trained supporting models for classification, image captioning, and video analysis to enhance visual search and power shoppable video- reduced manual catalog tagging effort by 50%.

Oct 2016 – Oct 2017 **AllGoVision – Software Engineer**

- Contributed to face recognition and OCR-based solutions, focusing on model fine-tuning models, preprocessing pipelines, and dataset creation for large-scale development.
- Fine-tuned VGG for face recognition on dataset on 3.4k identities - achieving 98.4% accuracy on the LFW benchmark.
 - Designed and implemented robust preprocessing pipeline data augmentation, outlier removal, class merging, and similarity filtering. That improved training efficiency and improved model accuracy.
 - Built OCR datasets for automatic number plate recognition (ANPR) and trained LeNet-based model for digit and character recognition - achieved 94% accuracy under real-world test conditions.

Projects & Solutions

E-Commerce & Fashion AI

Developed AI solutions for fashion e-commerce including clothing detection, visual similarity search, image classification, and automated captioning — enhanced product discovery, reduced manual catalog tagging, and improved retrieval relevance, enabling faster and more accurate catalog search.

Video Injection System — Shoppable Videos

Built end-to-end shoppable video pipeline with person and clothing detection, visual similarity search (triplet loss + FAISS), clustering, face recognition, and indexing — enabling interactive shopping experiences inside video content.

Video Understanding System

Designed large-scale video analysis system covering NSFW moderation, AI thumbnail generation, video classification, engagement prediction (I3D + VGGish), face recognition, scene/logo detection, and audio/video retrieval — automated moderation and improved recommendation accuracy.

Contextual Ads Injection System

Created contextual ad-injection engine powered by video understanding, text extraction, CLIP embeddings, and knowledge graph reasoning — delivering relevant ads that increased user engagement and ROI.

VisionScene and Movies X-Rays

Developed advanced video understanding tools for OTT platforms, including scene-level search, face identification, person tracking, and CLIP-based scene embeddings — enabling “X-Ray”-style interactive features for movies and series.

Skills & Data Expertise

Core Machine Learning & AI

Classification (Cross-Entropy, Triplet Loss), Object Detection (YOLO, FasterRCNN), Visual Similarity Search, Face Recognition, Person Tracking, Video Classification (I3D, Vggish), Content Moderation, OCR, Image Captioning, Sentiment Analysis, Multimodal Retrieval, RAG Systems, LLM Integration.

Frameworks & Tools

Python, PyTorch, TensorFlow, Keras, OpenCV, FAISS, LangChain, FastAPI, Kafka.

Deployment: Docker, AWS, GCP.

Other: Git, Data Preprocessing & Augmentation, Large-scale Data Curation.

Datasets & Data Creation

Face Recognition: Curated dataset of 3.4k identities (1k images each), outperforming open-source database available at the time; expanded to 10k identities for triplet-loss embedding training.

Fashion AI: Proprietary e-commerce catalog dataset for images, captions, and metadata for similarity search and classification; triplet-based data for shoppable video AI.

Video Moderation: Short-form video datasets annotated for NSFW, violence, unsafe objects, and audio-based content safety.

Video Engagement: Proprietary video dataset with watch-time, likes, and shares used as engagement signals; extracted I3D visual features and VGGish audio features.

VideoRAG & VisionScene: Multimodal datasets with video frames, scene captions, audio transcripts, and person/face tracking metadata for retrieval and search.